Quantifying Agricultural Resilience: A Geospatial Analysis of Drought, Soil, and the Mitigating Effect of Irrigation on Soybean Yield Disparity in Maryland

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Abstract

While the United States leads global soybean production, regional yield disparities persist, particularly in the Mid-Atlantic. Over the last 25 years, Maryland's mean soybean yield has risen to approximately 3,400–3,500 kg/ha; however, it consistently trails the U.S. national average. Between 2014 and 2024, Maryland's harvest-weighted mean yield (3,060 kg/ha) remained significantly below the national mean (3,350 kg/ha), matching or exceeding it in only two of those 11 years. This study investigates the environmental and management drivers of this disparity, focusing on the intersection of intensifying "flash drought" events, site-specific soil water-holding capacity, and irrigation infrastructure. We implemented a multi-sensor remote sensing approach using Sentinel-2 and Landsat data (2007–2024) to quantify the Normalized Difference Water Index (NDWI) during the critical R4–R6 reproductive window. Our results reveal an asymmetric vulnerability: the Southern Maryland district, characterized by low-buffering sandy loam (100 mm/m available water capacity), exhibits high yield volatility strongly correlated with drought severity ($r = 0.45$). In contrast, Maryland's Eastern Shore maintained physiological stability despite similar edaphic risks, with NDWI values remaining above metabolic stress thresholds due to extensive irrigation infrastructure. Irrigated sandy-loam counties achieved a mean yield of 2,990 kg/ha, a 260 kg/ha advantage over rainfed counterparts, while maintaining significantly narrower inter-annual yield spreads. These findings demonstrate that irrigation effectively decouples canopy water status from precipitation variability. Ultimately, this research highlights the necessity for data-driven, precision irrigation management in high-risk regions to stabilize production and bridge the persistent regional yield gap.

Keywords Soybean · Precision Agriculture · NDWI · Flash Drought · Google Earth Engine, Agricultural Districts, Mid-Atlantic

1. Introduction

The United States is a global leader in soybean production and the second-largest consumer, accounting for 32% of global soybean production (Skevas et al., 2025) and consuming 65.9 million tons annually (17.2% of global consumption) (Volkova & Smolyaninova, 2024). In the United States, the soybean value chain has an annual economic impact of $124 billion and supports more than 504,000 jobs (LMC International, 2023). In 2024, the average yield across the US was 3,409.6 kilograms per hectare (Schnitkey et al., 2025), a success largely attributable to the agroecological advantages of the Midwest's "Soybean Belt." In 2024, Maryland averaged 2,890 kilograms per hectare of yield (USDA NASS, 2025), a deficit of 520 kilograms per hectare in comparison to the national average. Historically (1997-2013), Maryland averaged 2,370 kilograms per hectare, compared to the national average of 2,720 kilograms per hectare (87.6% of the national average), consistently lagging by approximately 340 kilograms per hectare. While Maryland has posted impressive long-term growth (+34% from 1997-2001 to 2020-2024, slightly outpacing the national +32% gain), the state has never fully closed the yield gap, achieving parity with the national average in only 6 of the past 28 years, most recently in 2021 (USDA NASS, 2025). This disparity ~~is a byproduct of~~reflects environmental factors such as drought, precipitation, soil type, and irrigation.

Soybean productivity is profoundly sensitive to water deficits, a major stressor that frequently results in substantial annual yield reductions, sometimes reaching 50% under dry conditions (Arya et al., 2021; Peng et al., 2025). Research demonstrates that the timing and duration of water stress are critical factors determining the magnitude of yield loss (Dietz et al., 2021; Nguyen et al., 2023). Drought affects yield by altering physiological and molecular mechanisms, inhibiting biomass accumulation, disrupting symbiotic nitrogen fixation, and reducing photosynthetic capacity (Arya et al., 2021). Short periods of water stress during the growing period reduce both vegetative and generative yield (Dietz et al., 2021). In a long-term analysis across the US (1958-2007), meteorological drought accounted for an average of 13% of the variability in maize and soybean yields (Zipper et al., 2016). ¶

~~Furthermore, analysis~~Analysis in the US Midwest showed that climate variability suppressed average soybean yield gains by 30% during 1994-2013 (Mourtzinis et al., 2015). ~~Given this acute sensitivity, soybeans~~Soybeans are particularly susceptible when drought coincides with ~~their most vital growth phases.~~critical development periods of July to August (Zipper et al., 2016). Li et al. (2024) evaluated the impact of water deficit during early flowering stages on pod development stages (R1-R4) and documented seed yield reductions of up to 46%. ~~For soybean production, yield is most sensitive to short-term droughts (1-3 months) occurring during the critical development periods of July to August (Zipper et al., 2016). The reproductive development stages of soybean, particularly flowering and pod-setting (FPS), represent the most critical windows in the soybean's life cycle, where severe water deficit can cause devastating seed yield losses. Wei et al. (2018) reported a study conducted in China's Huabei Plain, in which severe water stress during the FPS stage caused losses of 73–82% per plant. Recent physiological research further demonstrates that this sensitivity is driven by a failure in "differential transpiration"; under combined heat and water stress, pods rely on open stomata to maintain internal temperatures up to 4°C cooler than the ambient air (Sinha et al., 2023). When water deficits inhibit this transpirational cooling, the resulting internal heat triggers embryo abortion, reduced seed filling, and permanent metabolic damage to the developing grain (Sinha et al., 2023).~~

~~The severe yield sensitivity of soybeans during reproductive development underscores the need to understand what controls field water availability.~~ Agricultural drought reflects a prolonged soil moisture deficit that often follows sustained precipitation shortfalls (Nguyen et al., 2023). Meteorological forces set drought in motion, but soil texture and hydraulic properties govern how severe and how long that deficit is at the root zone~~-~~—soil acts as the buffer between climate and crop response (Faé et al., 2020; Li et al., 2024). Clay loam generally confers strong drought resistance on soybeans because high clay content supports water retention (Li et al., 2024; Zhou et al., 2022); under drought, soybeans on clay loam showed the least physiological damage and the smallest rise in lipid peroxidation markers (MDA) relative to other textures (Zhou et al., 2022). ~~In Tennessee silty loam, available water capacity (AWC) can reach about 208 mm/m (Xie et al., 2024).~~ Despite ~~its~~ high water-holding capacity~~, research demonstrates that~~ in some soils, the inherent variability of precipitation in humid regions still requires supplemental irrigation to maximize yield (Singh et al., 2023; Xie et al., 2024). Sandy soils, due to their coarse texture, have low water retention and holding capacity, necessitating frequent irrigation (Suriadi et al., 2021; Zipper et al., 2016).~~ As a result, optimal production in tropical sandy loam soils necessitates more frequent irrigation to achieve the highest yields (Suriadi et al., 2021).~~

Technological advancements, such as remote sensing, provide a crucial avenue for deriving physiological metrics for the non-destructive assessment of plant status (Arya et al., 2021). ~~The effectiveness of remote sensing is often categorized by the specific wavelengths employed, distinguishing indices based on crop health from those based on water content. For instance, the~~The Normalized Difference Vegetation Index (NDVI), calculated as the normalized ratio of the difference between near-infrared (NIR) and red light (R) reflectance to their sum, is a common indicator of vegetation cover, health, and crop growth status (Jackson et al., 2024).

According to Basal and Szabó (2020), drought-stressed treatments exhibited a 35.5% mean decrease in NDVI compared to non-stressed conditions. However, NDVI is ~~considered~~ limited for precise estimation of vegetation water content (VWC) because it ~~is more sensitive to chlorophyll content and~~ saturates under dense vegetation cover~~, failing to capture the full range of VWC changes~~ (Jackson et al., 2024; Tian et al., 2025). ~~Because of this, the~~The Normalized Difference Water Index (NDWI) ~~is heavily used, leveraging the~~leverages strong water absorption features ~~found~~ in short-wave infrared (SWIR) bands (1200–2500 nm), ~~which makes~~making it ~~demonstrably~~ better than NDVI for VWC estimation in crops like soybean (Gao et al., 2015; Jackson et al., 2024; Xu et al., 2020). ~~NDWI contrasts the reflectance of the NIR channel with that of the SWIR channel, which is effective because SWIR wavelengths (1200–2500 nm) exhibit strong water absorption features and are highly sensitive to leaf water content (Xu et al., 2020). Xu et al. (2020) found high correlations between NDWI and soybean VWC using a combination of MODIS and Landsat data, resulting in an R^2 value of 0.85 for canopy VWC estimation in Iowa. While vegetation indices like NDWI accurately diagnose the physiological consequences of water deficit within the plant canopy, they provide only a snapshot of the current stress status. To contextualize this real-time plant response and model the broader environmental drivers influencing long-term stress and yield, meteorological drought indices are essential. The calculation of the Palmer Drought Severity Index (PDSI) relies on an implicit soil water balance model, requiring inputs of precipitation, temperature, and the area's specific soil water-holding capacity. A study in Iowa found a mean correlation coefficient of 0.46 between PDSI and soybean yields, indicating a strong association over time (Islam et al., 2025). In summary, the combined use of physiological indices like NDWI and meteorological indices like PDSI provides a comprehensive, multi-modal framework necessary for the accurate diagnosis of~~

~~water stress and the subsequent implementation of optimized precision agricultural strategies.~~ Xu et al. (2020) found high correlations (R^2 = 0.85) between NDWI and soybean VWC using MODIS and Landsat data in Iowa.

~~The necessity of irrigation as a mitigation strategy against drought and water stress is increasingly critical, irrespective of inherent soil characteristics, particularly given the intensifying threats posed by climate variability. Soybeans are a water-intensive crop, and their production is~~Soybeans are a water-intensive crop highly dependent on optimum rainfall or sufficient irrigation (Arya et al., 2021). In Tennessee, Xie et al. (2024) reported irrigated yields up to ~30% above rainfed over five years~~, illustrating how irrigation improves yield stability and raises production under variable rainfall.~~. In an extremely dry year in the Mississippi Delta, non-irrigated crops yielded far less than irrigated crops (Zipper et al., 2016). ~~The goal of modern irrigation management is to optimize the timing and amount of water applied, thereby maximizing Irrigation Water Productivity (IWP) while achieving the highest yield.~~ Supplemental irrigation ~~is recommended~~ during ~~the~~ reproductive stages (R1–R8) ~~to enhance~~enhances soybean growth and yield substantially (Li et al., 2024). Remote sensing tools~~, such as~~ like NDWI~~, can accurately identify critical reproductive stages, including flowering (R1 and R2) and the onset of seed filling (R5), enabling~~ enable precise~~, timely~~ irrigation scheduling by identifying critical stages (Braga et al., 2021). Overall, irrigation ~~is indispensable for stabilizing and maximizing soybean~~stabilizes production amid climate change~~, and integrating advanced monitoring tools provides the technological framework necessary to apply water precisely when and where the crop needs it most, regardless of intrinsic soil-type limitations~~ (Miller et al., 2018).

While general principles connect climate and irrigation to yield, regional research in Maryland ~~has uniquely focused on integrating~~ integrates these variables to address ~~the region's~~ intensifying climate variability and aquifer strain. Recent ~~investigations at the~~ Beltsville Agricultural Research Center (BARC) ~~have demonstrated~~investigations demonstrate the need ~~to integrate~~for physical data with remote sensing to address yield heterogeneity. Chang et al. (2025) found ~~that while~~ hydro-topographic models ~~alone~~ explained ~~less than~~ <40% of spatial yield variation, but incorporating hydrological factors and ~~remote-sensing data (~~NDVI~~)~~ substantially increased explanatory power~~, underscoring that characterizing the environmental drivers of yield in Maryland with high-resolution spectral data is a sound strategy. Concurrently, broader regional studies have addressed the resource implications of this variability, noting that groundwater~~. Groundwater extraction for irrigation surged ~~by approximately~~ ~38% ~~between~~ (2000 ~~and~~ -2015~~,~~), driving ~~declines in~~ Coastal Plain ~~aquifers~~ aquifer declines (Paul et al., 2021). Rahman et al. (2024) ~~conducted a modeling study in Maryland using the Soil and Water Assessment Tool (~~used SWAT~~)~~ and ~~the~~ IrrigEcon ~~decision support system to assess water productivity and economic returns, demonstrating that~~to show supplemental irrigation ~~can increase crop~~increases yields ~~by approximately~~ 110–130% ~~during~~in dry years while remaining economically viable. Projections confirm Peninsula (Eastern Shore) soybean net irrigation requirements exceed Mainland by 50-70 mm/season baseline, rising modestly to 945-964 mm RCP8.5 (2040-2100) due to ET0 dominance (r=0.92) over precipitation (Borzì & Kumar, 2026)

However, existing ~~regional~~ research ~~tends to focus either~~focuses on micro-scale hydro-~~topographic processes~~topography (Chang et al., 2025) or ~~on modeling~~ regional water policy ~~and economic consequences~~/economics (Paul et al., 2021; Rahman et al., 2024). ~~A considerable~~A gap remains in ~~bridging these scales, as existing studies are limited in scope and rarely focus on~~ field-scale soybean production, ~~or the specific driving factors of yield loss unique to Maryland. Furthermore, no published framework currently leverages the complementary~~

~~strengths of~~ integrating meteorological drought indices (PDSI) and physiological indices (NDWI) to identify ~~regional~~ irrigation priorities. Without such a unified model ~~that integrates long-term environmental stress with real-time water deficits~~, Maryland growers lack ~~the diagnostic~~ tools ~~necessary to implement optimized~~to optimize irrigation ~~practices~~ and close the ~~state's persistent~~ yield gap.

Therefore, this study establishes a multi-modal diagnostic framework ~~that combines~~combining meteorological indices with high-resolution physiological data to quantify ~~the relationship between~~ irrigation ~~and~~ -soybean ~~yields~~yield relationships in Maryland. By mapping ~~where~~ NDWI ~~falls within metabolic~~ stress bands associated with foregone yield, we provide an empirical basis for ~~targeting~~targeted water management ~~and agronomic interventions~~ to narrow ~~the state's~~Maryland's persistent yield gap.

2. Materials and Methods

2.1 Study Area Description

For statistical and agricultural purposes, Maryland is divided into five USDA National Agricultural Statistics Service (NASS) reporting districts that align with the state's major physiographic and farming regions: Western, North Central, Southern, Upper Eastern Shore, and Lower Eastern Shore (Fig. 1).

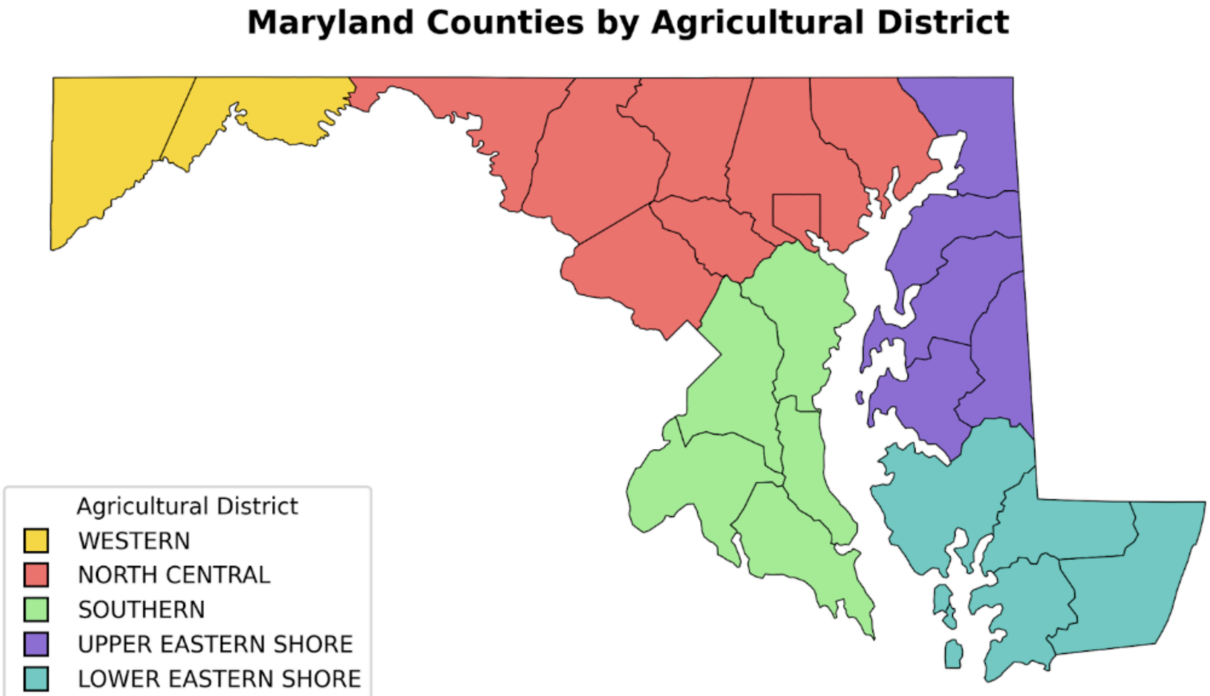

Figure 1: Maryland counties categorized by USDA NASS Agricultural Reporting Districts.

The Western district, which comprises the Appalachian highlands, includes Allegany and Garrett Counties. The North Central district, which largely encompasses Piedmont and metro-fringe cropland, includes Baltimore City, Carroll, Frederick, Harford, Howard, Montgomery, and Washington Counties. The Southern district, located entirely within the western Coastal Plain, includes Anne Arundel, Calvert, Charles, Prince George's, and St. Mary's Counties. The Upper Eastern Shore district, which covers the northern Delmarva Peninsula, encompasses Caroline, Cecil, Kent, Queen Anne's, and Talbot Counties. Finally, the Lower Eastern Shore district, covering the southern Delmarva Peninsula, encompasses Dorchester, Somerset, Wicomico, and Worcester Counties. These districts are the standard units used by NASS for data aggregation and trend analysis, reflecting meaningful spatial variations in climate, edaphic properties, and production systems across the state (USDA NASS, 2025).

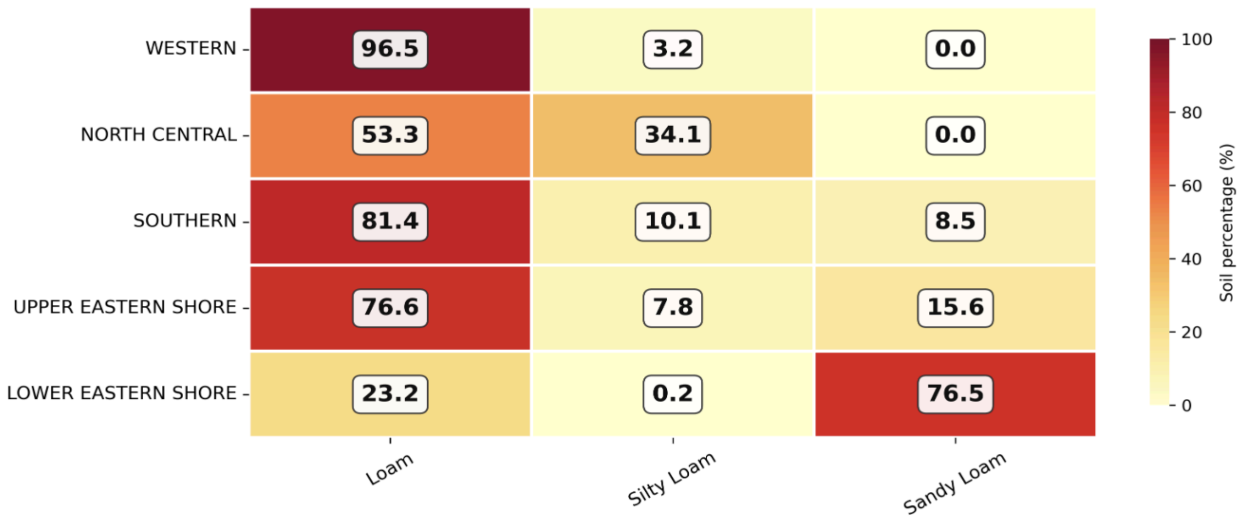

Figure 2: Average soybean soil composition by agricultural district (2020).

Maryland's reporting districts are characterized by a profound physiographic and pedological divide established along the Fall Line, which separates the crystalline rocks of the Piedmont Plateau from the unconsolidated sediments of the Atlantic Coastal Plain. Loam is a primary soil constituent across most of the state (Fig. 2), yet this underlying geology dictates its agricultural performance. Although this district possesses the state's highest concentration of Loam at 96.5% (Fig. 2), its AWC of 140.00 millimeters per meter (Klopp & Bly, 2024) provides only a moderate moisture reservoir. When coupled with steeper topography and a highland environment, this results in distinct climatic conditions, including a mean annual precipitation of 114.6 cm and lower precipitation during the critical growing season than in the rest of the state (Fig. 3).

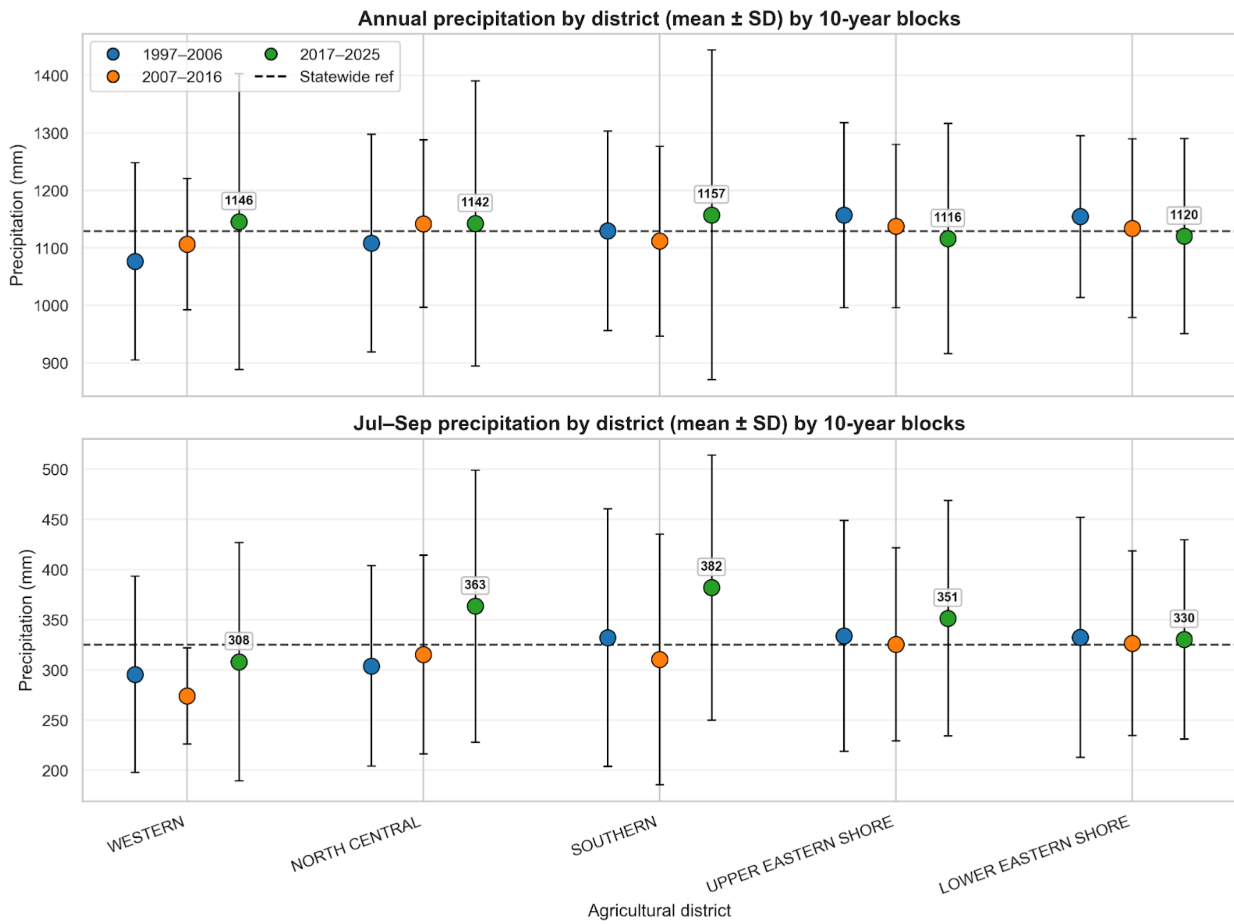

Figure 3: Annual and growing season (Jul–Sep) precipitation trends by district across 10-year blocks (1997–2025).

The contrast between the North Central and Southern districts highlights how similar climatic inputs can produce different hydrological environments due to differences in soil buffering. Between July and September, both districts receive comparable mean precipitation, 363 mm and 382 mm, respectively (Fig. 3). However, their ability to store water for crop use differs markedly. The North Central district benefits from the state's highest concentration of Silty Loam (34.1%), a texture characterized by intermediate porosity that retains high volumes of water while maintaining necessary root aeration (Fig. 2). For context, Tennessee silty loam achieves available water capacity (AWC) of ~208 mm/m (Xie et al., 2024), while Southern Maryland's sandy loam fraction limits AWC to ~100 mm/m (Klopp & Bly, 2024). This ~50% reduction in moisture reservoir heightens Southern district sensitivity to rainfall timing, despite higher precipitation inputs. These soils possess a high Plant AWC, the “moisture reservoir” held between field capacity and the permanent wilting point, of approximately 200 mm/m of depth (Klopp & Bly, 2024).

In contrast, the Southern district is 81.4% Loam but includes substantial Sandy Loam (8.5%) (Fig. 2), which has an AWC of only 10.0 cm per meter (Klopp & Bly, 2024). This 50% reduction in the moisture reservoir makes Southern Maryland crops highly sensitive to the specific timing of summer rain, as the soil lacks the natural buffering capacity found in the Piedmont. The Upper and Lower Eastern Shore districts represent a “hydrological paradox” within this spectrum. These regions have the highest concentrations of Sandy Loam in the state, peaking at 76.5% in the Lower Eastern Shore (Fig. 2). While this would normally imply a low AWC of 100 mm/m, the Eastern Shore maintains physiological stability through extensive irrigation enabled by hydrogeologic accessibility. This is enabled by the Surficial aquifer, in which ancient Quaternary paleochannels yield transmissivity rates ranging from 47.38 to 4,970.31 meters squared per day (Maryland Geological Survey, 2026). While the Southern district's upland aquifers are thinner and more fragmented, forcing a reliance on a natural soil buffer of 100–140 mm/m, the Eastern Shore’s unique hydrogeologic conduits allow it to bypass these pedological limits entirely.

2.2 Data Acquisition and Analysis

The primary datasets for this study were compiled from US agricultural and meteorological databases to investigate the relationships among yield, irrigation, and environmental stress. The computational analysis was performed using a standardized Python environment. Primary agricultural production and irrigation datasets were acquired from the USDA National Agricultural Statistics Service (NASS) (USDA NASS, 2025). County-level soybean yield and production statistics were extracted using the Quick Stats Ad-hoc Query Tool, enabling longitudinal analysis of productivity across Maryland's five reporting districts. To differentiate between rainfed and managed hydrological systems, data regarding irrigated soybean acreage were sourced from the USDA Census of Agriculture. Unlike the annual NASS surveys, which utilize representative sampling to estimate yearly yields, the Census of Agriculture provides a comprehensive survey of the entire producer population at five-year intervals (e.g., 2017, 2022). This high-resolution dataset established the baseline for irrigation density within the Atlantic Coastal Plain and enabled characterization of districts by the extent of managed water infrastructure.

The environmental context for the July-September study window was established using data from the National Centers for Environmental Information (NCEI) "Climate at a Glance" system (NOAA, 2026). Monthly precipitation totals were collected to establish seasonal moisture regimes for each reporting district.

To capture the nuances of moisture stress during the core reproductive stages of the soybean crop, three distinct Palmer drought metrics were utilized:

- Palmer Drought Severity Index (PDSI): Employed to monitor meteorological drought and short-term moisture departures, reflecting immediate water availability at the root zone.
- Palmer Hydrological Drought Index (PHDI): Utilized to assess long-term moisture impacts on groundwater and reservoir levels. This is critical for evaluating the stability of the Eastern Shore’s Surficial aquifer.
- Palmer Modified Drought Index (PMDI): Provided a balanced metric reflecting the transition between meteorological and hydrological drought conditions, ensuring a comprehensive assessment of environmental stress.

Meteorological drought indices like the Palmer Drought Severity Index (PDSI) provide essential context for NDWI snapshots. PDSI calculation uses an implicit soil water balance model with precipitation, temperature, and site-specific soil water-holding capacity inputs. In Iowa soybeans, PDSI showed a mean $r = 0.46$ correlation with yields (Islam et al., 2025), validating its linkage to production impacts observed here.

To capture a continuous record of soybean phenology from 2007 to 2024, a multi-sensor harmonized approach was implemented. For the modern period (2017–2024), the primary data source was the Sentinel-2 Level-2A (L2A) surface reflectance collection (European Space Agency, 2024), which provides Bottom-of-Atmosphere (BOA) reflectance at 10-meter to 20-meter resolution. For the 2008–2016 period, the study used Landsat 8 (C02/T1_L2) (U.S. Geological Survey, 2016) and Landsat 7 data to maintain temporal coverage during periods when Sentinel-2 data were unavailable.

To mitigate the impact of frequent cloud cover common in agricultural monitoring, pixels were filtered using the QA60 bitmask and the S2_CLOUD_PROBABILITY dataset. Applying a cloud probability threshold of >35% effectively balances data availability with spectral purity in multi-temporal composites (Huang et al., 2024). To overcome residual atmospheric noise and "mixed pixel" interference, 10-day median composites were generated. These composites typically retain the median spectral value across the time window, a process critical for capturing rapid phenological transitions, such as the shift from flowering (R1/R2) to pod development (R3/R4).

Biophysical analysis was conducted by calculating key vegetation and moisture indices for every pixel using the visible, Near-Infrared (NIR), and Short-Wave Infrared (SWIR) bands. The

Normalized Difference Vegetation Index (NDVI) was used to monitor chlorophyll activity and biomass:

$$NDVI = (\rho NIR - \rho RED) / (\rho NIR + \rho RED)$$

where ρNIR and ρRED correspond to Sentinel-2 Bands 8 and 4, respectively; however, because NDVI is prone to saturation during the high-biomass R4–R6 stages (Barboza et al., 2023), the Normalized Difference Water Index (NDWI) was also calculated. NDWI directly measures leaf water content and remains sensitive to moisture stress even in dense canopies:

$$NDWI = (\rho GREEN - \rho NIR) / (\rho GREEN + \rho NIR)$$

NDWI ~~This~~ was calculated using Sentinel-2 Band 3 (Green) and Band 8 (NIR), leveraging SWIR water absorption (1200–2500 nm) for sensitivity to leaf water content even in dense canopies. NDWI contrasts NIR reflectance with SWIR, where SWIR exhibits strong water absorption features highly sensitive to vegetation water content (Xu et al., 2020). To isolate soybean fields, the USDA Cropland Data Layer (CDL) (USDA NASS, 2024) was used as a mask. All Google Earth Engine (GEE) spatial operations utilized U.S. Census TIGER boundaries: state polygons from TIGER/2018/States and county polygons from TIGER/2016/Counties (U.S. Census Bureau, 2018).

3. Results and Discussion

3.1 Regional Physiography and Dynamics

Soybean yields in Maryland increased significantly over the last 25 years, a trend confirmed by a paired analysis of county-level data. A paired t-test comparing mean yields from the 1997–2006 window (2,294.5 kg/ha) to the 2019–2025 window (3,077.7 kg/ha) revealed a highly significant average increase of 783.2 kg/ha ($p < 0.001$). This robust upward trajectory was further validated by a Wilcoxon signed-rank test ($p < 0.001$), ensuring that the growth reflected a consistent regional shift rather than the influence of a few high-yielding outliers. This historical trajectory reflected a broader national trend where U.S. yields doubled from 1,614.0 kg/ha in 1960 to over 3,429.8 kg/ha by 2020 (USDA NASS, 2025).

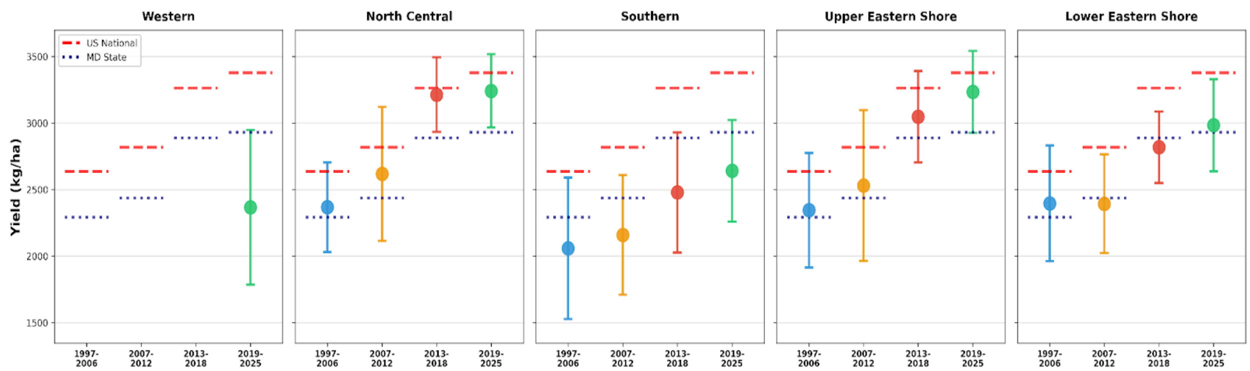

Figure 4: Historical soybean yield (kg/ha) trends by agricultural district across four distinct time periods (1997–2025). Each panel shows the mean yield for a reporting district, with error bars representing the standard deviation within that 10-year block. The red dashed line denotes the U.S. national yield benchmark for the corresponding period, while the navy dotted line indicates the Maryland state average.

Despite these advancements, a consistent “yield gap” persisted between Maryland’s performance and the U.S. national average. This disparity was acutely visible across Maryland's agricultural districts. The Eastern Shore districts tended to perform relatively well (Fig. 4), particularly the Upper Eastern Shore. Counties such as Queen Anne's and Talbot represent the state's most intensive production areas; during the study period, they consistently matched or exceeded state averages by benefiting from silt loam soils with high AWC and “sub-irrigation” effects from shallow water tables. The Eastern Shore’s advantage is further amplified by its proximity to the shallow water tables of the Columbia Aquifer, which can contribute between 6.5% and 24% of the total soybean water requirement through capillary rise (Paul et al., 2021). Although the Lower Eastern Shore followed a similar growth trajectory, it is characterized by predominantly sandy loam soil (Fig. 2), which adversely affected yield relative to the Upper Eastern Shore.

The complex topographic architecture of the rolling Piedmont plateau primarily influences soybean yield in the North Central district. Unlike the relatively uniform Coastal Plain of the Eastern Shore, field-scale yield in this region was directly influenced by landscape position, which governs the dynamic redistribution of soil moisture and organic matter. Landscape features alone were shown to account for between 6% and 48% of the variation in soybean yield (Chang et al., 2025). Despite this topographic variability, the North Central district performed well overall because it received the second-highest volume of growing-season precipitation (Fig. 3) while maintaining the state's highest concentration of silty loam (Fig. 2), which has an AWC of 200 mm/m. In contrast, while the Southern district received the highest overall growing-season precipitation (Fig. 3), its productivity was hindered by a higher percentage of sandy loam and a lack of the silty loam found in the North Central region (Fig. 2). Consequently, this resulted in lower overall yields (Fig. 4) relative to the North Central district.

While NASS data on Western Maryland soybean production became available only in 2021 (primarily from Garrett County), the records indicated high volatility. A primary driver for this yield instability is the rugged topography of the Western Piedmont and Appalachians. As noted by Chang et al. (2025), higher landscape positions, such as hilltops and shoulders, are characterized by higher runoff, thinner A horizons, and lower water availability, leading to the chronic underperformance observed in rainfed years (Chang et al., 2025).

3.2 Drought Dynamics and Yield Decoupling

The Maryland agricultural landscape underwent a regime shift from slow-onset moisture deficits toward “flash droughts”, rapid-onset events that reach peak severity within a narrow four-week window (Chen et al., 2019). While historical benchmarks occurred in 1998, 2006, and 2010, the most recent five-year period (2021–2025) indicated a heightened cluster of activity (Fig. 5). In

2024 and 2025, approximately 20% of Maryland counties were affected by flash droughts, a notable increase compared to the relative stability observed between 2013 and 2020 (Fig. 5). These events are particularly devastating when they coincide with the critical R4 to R6 stages, leading to disproportionate yield losses relative to the magnitude of total seasonal precipitation deficits.

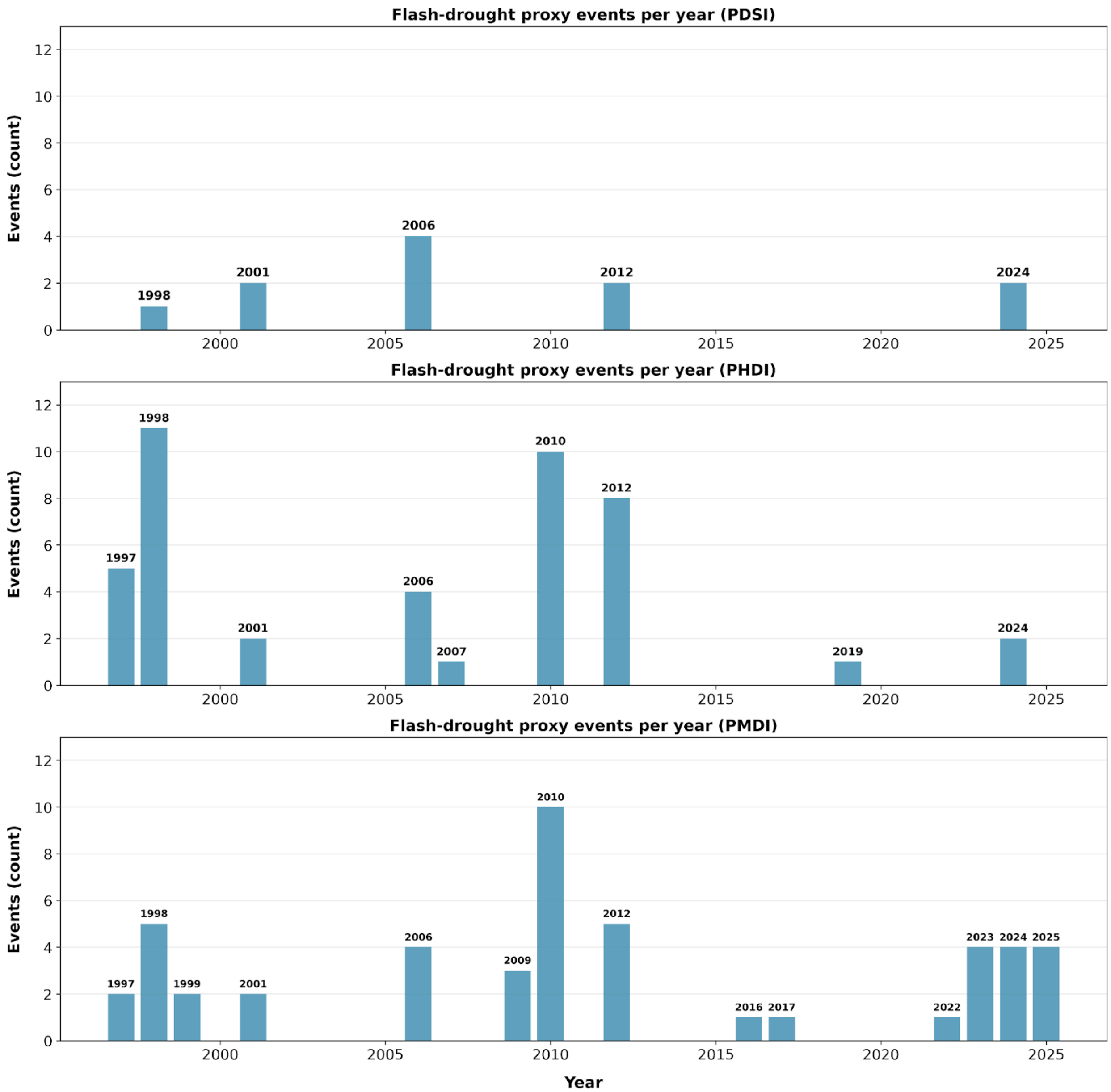

Figure 5: Annual counts of flash-drought proxy events by Palmer index (PDSI, PHDI, PMDI) for Maryland counties. Events satisfy a monthly proxy: drought severity worsens by at least two

categories within one month (~4 weeks), the Palmer value does not increase over that step, and the peak severity is sustained at least one additional month.

For optimal soybean production, a critical “cut-off point” is reached when soil moisture levels fall to 50% of the crop water requirement (CWR). Beyond this threshold, moisture limitation severely reduces stomatal conductance and sub-stomatal carbon dioxide concentration, ultimately suppressing photosynthetic rates and grain yield (Mwamlima et al., 2022). Despite similar edaphic risks, yield outcomes varied sharply across districts due to management disparities. The Southern district demonstrated the highest volatility in response to drought severity; as illustrated in Fig. 6, it exhibited a strong Pearson correlation ($r = 0.45$) between yield and drought intensity, with “Severe” drought years resulting in a dramatic yield collapse.

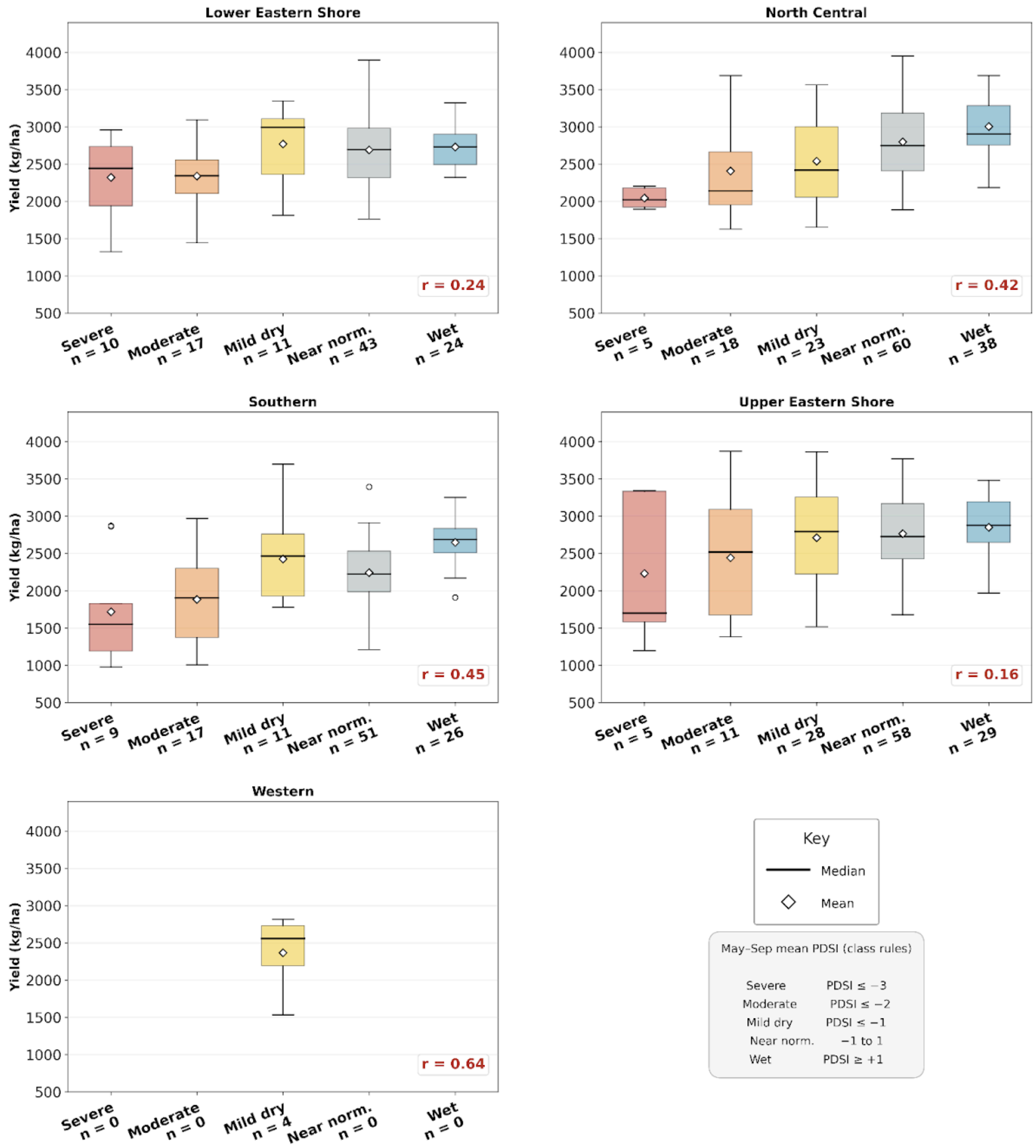

Figure 6: County-year soybean yield (kg/ha) by May–September mean Palmer Drought Severity Index (PDSI) drought class, shown for each Maryland agricultural district. Boxplots summarize yields within each class; n is the number of county-year observations per box.

In contrast, the Upper and Lower Eastern Shore districts exhibited greater resilience, even when encountering similar sandy loam profiles. This decoupling of yield from climatological drought was largely attributed to the intensification of irrigation in these regions. Harvested irrigated soybean acreage surged since 1997, with the Upper Eastern Shore reaching 9,854 hectares and the Lower Eastern Shore totaling about 8,884 hectares by 2022 (Fig. 7). This contrasted sharply with the Southern district, where irrigated data was frequently suppressed or negligible, leaving its producers entirely reliant on rainfall and conservation agriculture to maintain the “hydraulic floor.”

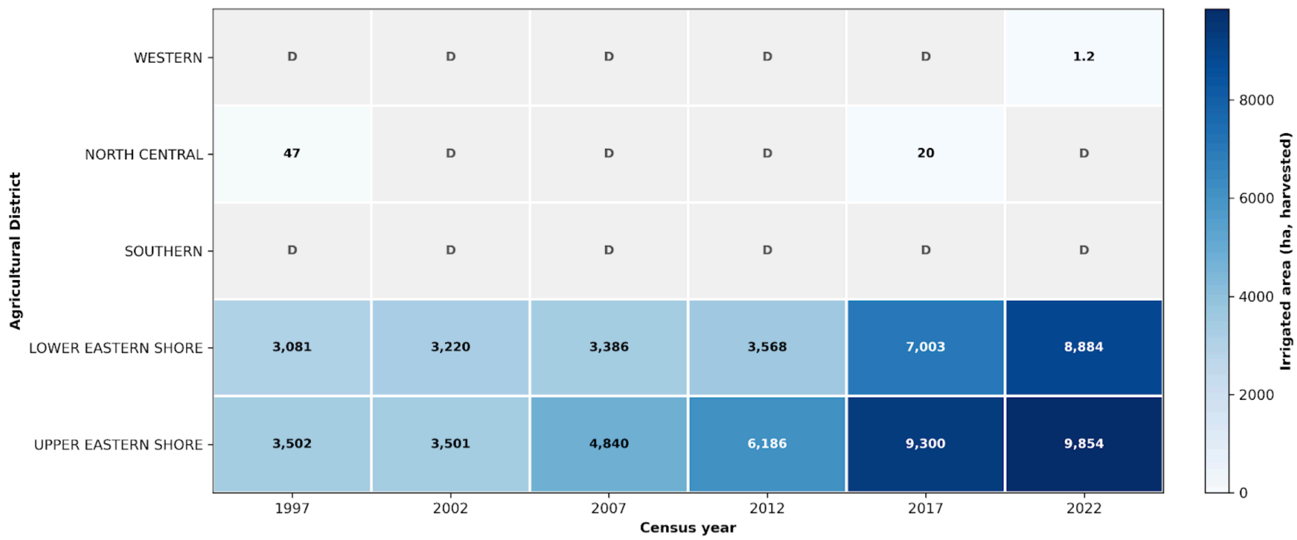

Figure 7: Longitudinal trends in irrigated soybean area (ha) by Maryland agricultural district (1997-2022). Values represent the sum of harvested irrigated hectares for all counties within each USDA NASS reporting district per census year. Cells marked “(D)” indicate data suppressed at the county level by NASS to avoid disclosing data for individual operations.

3.3 Biophysical Response and Stress Dynamics

The biophysical response of Maryland soybeans to flash drought is best characterized by canopy “greenness” and internal water status, as quantified through NDVI and NDWI. A comparison of NDWI distributions across Maryland districts reinforced the critical role of irrigation in maintaining optimal physiological conditions. In the South Fork Experimental Watershed (Iowa), NDWI values during the critical reproductive growth stage were observed to range from 0.37 to 0.58, corresponding to high-productivity yields exceeding 4,035 kg/ha (Xu et al., 2020). Conversely, experimental soybean plants under induced water deficit conditions exhibited NDWI values ranging from 0.121 to 0.144 (Braga et al., 2021). Lower NDWI values were consistently associated with reduced leaf water content, signifying metabolic stress.

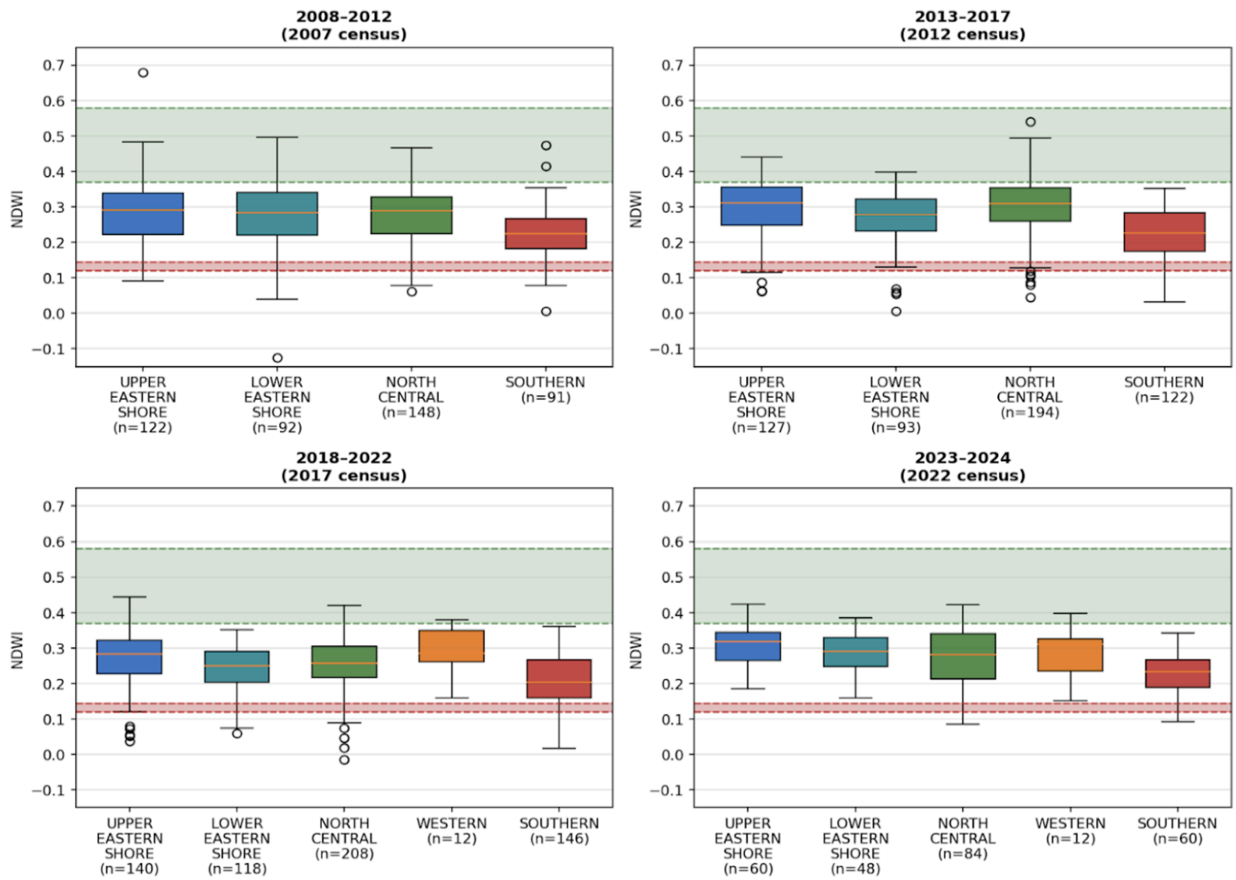

Figure 8: Spatiotemporal distribution of NDWI values by Maryland agricultural district and census period during the R4–R6 reproductive window (DOY 200–260). Boxplots represent 10-day median values. Green and red shaded regions indicate optimal productivity (0.37–0.58) and metabolic stress (0.121–0.144) thresholds, respectively. Districts are included based on USDA NASS yield data availability.

This sensitivity during reproductive stages is driven by physiological mechanisms. The reproductive development stages of soybean, particularly flowering and pod-setting (FPS), represent the most critical windows in the soybean's life cycle, where severe water deficit can cause devastating seed yield losses. Wei et al. (2018) reported severe water stress during FPS caused losses of 73–82% per plant in China's Huabei Plain. Recent research demonstrates failure in "differential transpiration" under combined heat/water stress: pods rely on open stomata to maintain internal temperatures up to 4°C cooler than ambient air (Sinha et al., 2023). Water deficits inhibit transpirational cooling, triggering embryo abortion, reduced seed filling, and permanent metabolic damage (Sinha et al., 2023). Lower NDWI values consistently correspond to these metabolic stress thresholds (Braga et al., 2021)

The distribution of NDWI during the R4–R6 window (DOY 200–260) across Maryland's agricultural districts and census periods confirms a persistent moisture deficit in the Southern district (Fig. 8). As shown in the boxplot distributions, the Southern district consistently maintained the lowest median NDWI and the highest frequency of observations within the critical metabolic stress zone (0.121–0.144), a range associated with yield-limiting water deficits (Braga et al., 2021). Across the four longitudinal periods (2008–2024), Southern Maryland's median NDWI ranged from 0.20 to 0.23. Notably, nearly 100% of these observations fell below the optimal productivity band (0.37–0.58) defined by Xu et al. (2020), highlighting a chronic departure from high-yield environmental conditions.

Across all R4–R6 NDWI observations (DOY 200–260), district means were not equal: a one-way ANOVA showed big differences among the five districts ($p < 0.001$), and a Kruskal–Wallis test on the same groups was consistent ($p < 0.001$). Focused comparisons then showed that the mean NDWI was lower in the Southern district than in the Upper Eastern Shore ($p < 0.001$) and Lower Eastern Shore ($p < 0.001$). The share of observations in the stress band (0.121–0.144) was also higher in the Southern district than in the pooled Eastern Shore ($p <= 0.002$). This pattern occurred even though the Eastern Shore has a higher share of sandy loam and lower AWC, which supported the interpretation that supplemental irrigation helped keep canopy water status above roughly 50% AWC, limiting pod abortion and metabolic shutdown.

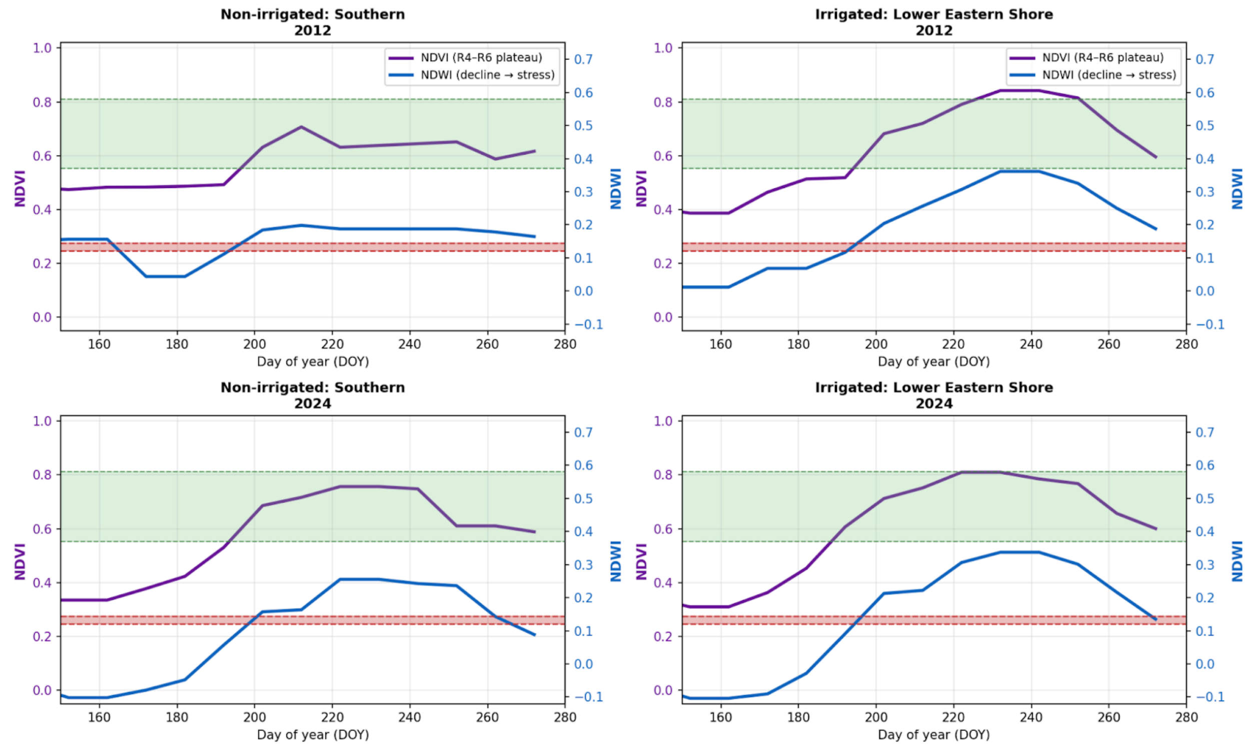

Figure 9: Dual-axis time series of NDVI and NDWI illustrating the "Invisible Stress Window" during the R4–R6 reproductive stages. The left axis represents NDVI, while the right axis measures NDWI. The green shaded region denotes the optimal NDWI range (0.37–0.58) observed in high-yielding reference regions.

The dual-axis comparison (Fig. 9) illustrates that the Lower Eastern Shore maintained superior canopy water status despite sandier soil profiles and lower 2024 precipitation (1,146 mm) relative to the Southern district (1,251 mm). During the 2024 reproductive window, the rainfed Southern district's median NDWI (0.143) fell directly into the stress band, with 54% of 10-day composites remaining at or below the 0.144 threshold. Conversely, the Lower Eastern Shore remained resilient despite considerably less annual rainfall. The data indicates that supplemental irrigation infrastructure on the Lower Eastern Shore (Fig. 7) effectively decouples canopy-level water status from both precipitation and soil AWC. This mechanism maintains leaf water content and photosynthetic capacity above critical stress thresholds, explaining the more favorable biophysical profiles observed in the Lower Eastern Shore compared to the higher-precipitation, yet moisture-vulnerable, Southern district.

3.4 Irrigation and the Yield Stability Gap

The preceding analysis demonstrated that irrigation decouples soybean canopy water status from both precipitation and soil water-holding capacity, yet the spatial distribution of irrigation in Maryland remains sharply uneven. As of the 2022 Census of Agriculture, the Upper Eastern Shore reported 9,854 hectares and the Lower Eastern Shore reported 8,884 hectares, while the Southern district's irrigated acreage remained suppressed or negligible across every census year from 1997 through 2022 (Fig. 7). Among the 11 Maryland counties where sandy loam exceeds 8% of the soil profile, the six irrigated counties (Caroline, Dorchester, Queen Anne's, Talbot, Wicomico, and Worcester) averaged approximately 37,800 planted soybean acres. In contrast, the five rainfed counties (Anne Arundel, Calvert, Charles, Prince George's, and St. Mary's) averaged only 9,100 acres (Fig. 10). This disparity suggested that irrigation infrastructure is currently economically optimized only where production scale is sufficient to amortize the capital costs of deep wells and center-pivot systems.

The goal of modern irrigation management is to optimize timing and amount applied, maximizing Irrigation Water Productivity (IWP) while achieving highest yields. NDWI precisely identifies critical reproductive stages—flowering (R1/R2) and seed filling onset (R5)—enabling timely scheduling that maintains canopy water above stress thresholds even on sandy loam (Braga et al., 2021; Li et al., 2024)

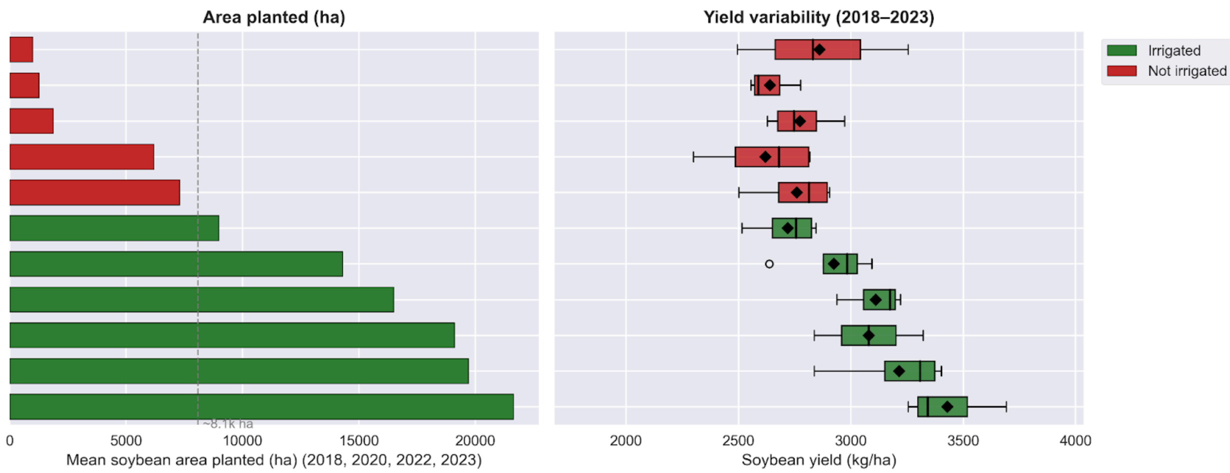

Figure 10: Soybean production and yield stability in Maryland counties with > 8% sandy loam. Panels show mean planted acreage (2018–2023; left) and yield distributions (kg/ha; right). Boxplots indicate IQR, median (line), and mean (diamond). Counties are ranked by descending acreage; colors denote irrigated (green) or rainfed (red) systems.

The yield consequences of this infrastructure gap extended beyond mean performance to inter-annual stability. Across the 2018–2023 survey years, irrigated sandy loam counties averaged ~2,993 kg/ha compared with ~2,730 kg/ha in rainfed counterparts, a baseline gap of ~260 kg/ha. However, Fig. 10 illustrates that rainfed counties exhibited substantially wider yield distributions; for example, St. Mary's County yields fluctuated between ~2,287 kg/ha and ~2,825 kg/ha, while irrigated counties such as Queen Anne's remained tightly clustered between ~3,228 kg/ha and ~3,700 kg/ha. This yield compression reflected the buffering effect of supplemental water, which limits the catastrophic low-yield outcomes seen in moisture-deficit years. In extreme years, the gap widened; in 2024, the Southern district yielded ~2,091 kg/ha while the irrigated Eastern Shore districts reached ~3,067 kg/ha, a deficit of ~976 kg/ha.

Future NA-CORDEX projections validate Peninsula resilience: NIR remains 50-70 mm/season higher than Mainland through 2100 (873-950 mm baseline Peninsula), but modest intensification (945 mm RCP8.5 mid-century) suggests infrastructure viability absent aquifer collapse (Borzì & Kumar, 2026). ¶

A critical finding of this study was that the regions most vulnerable to flash drought were precisely those with the least irrigation coverage. While the Southern district receives the highest growing-season precipitation (Fig. 3), its yield remained the lowest and most volatile because its sandy loam fraction reduces effective water storage, and it lacks the managed hydrological floor present on the Eastern Shore. This created an asymmetric risk profile: Southern producers absorbed the full impact of flash drought events during the R4–R6 window, while Eastern Shore producers with comparable edaphic limitations mitigated that risk through irrigation. As flash drought frequency increased (Fig. 5), the yield penalty for non-irrigation continued to grow. Expanding irrigation access in the Southern district, through cost-share programs or precision micro-irrigation suited to smaller operations, could narrow the persistent state-wide yield gap, compress the volatility visible in Fig. 10, and bring rainfed counties closer to the production ceiling demonstrated by the irrigated Eastern Shore.

4. Conclusion

Maryland soybean yields have demonstrated a consistent upward trajectory over the past 25 years. However, a significant yield gap and uneven resilience to environmental water stress remain. This study indicates that the Southern district continues to lag behind national benchmarks despite high seasonal precipitation, a consequence of low-buffering sandy loam soils and a lack of irrigation infrastructure. Conversely, the Eastern Shore benefits from favorable sub-irrigation and widespread supplemental irrigation, which decouple canopy water status from meteorological deficits.

As the frequency of flash drought events increases, particularly during the critical R4–R6 reproductive window, the biophysical evidence from NDVI and NDWI confirms that irrigated districts maintain superior canopy water status and tighter yield distributions compared to rainfed regions. Currently, irrigation in Maryland is concentrated where planted acreage is large (Fig. 10), leaving the most vulnerable district, Southern Maryland, with the minimal buffer. Closing the Maryland yield gap requires expanding irrigation access in underserved, high-sandy loam areas such as Southern Maryland. Integrating advanced monitoring tools like NDWI provides the technological framework necessary to apply water precisely when and where the crop needs it most. Expanding access through cost-share programs, community systems, or precision micro-irrigation should be a regional priority to reduce year-to-year volatility and strengthen the long-term resilience of Maryland soybean production.

CRediT authorship contribution statement

Ananth Sriram: Data curation, Formal analysis, Investigation, Visualization, Writing – original draft. **Hemendra Kumar**: Conceptualization, Methodology, Supervision, Writing – review & editing, Funding acquisition. **Iolanda Borzì**: Investigation, Writing – review & editing

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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